

■ ARDUINO KEYPAD PASSWORD LOCK PROJECT

Lesson Notes & Reference Guide

■ Project Build a keypad-operated password lock with servo	■ Components Arduino Uno, 4×4 Keypad, Servo Motor, Jumper Wires	■ Skills Gained Wiring, coding, password logic, troubleshooting
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What is Arduino?

Arduino is a small, programmable computer on a circuit board. It acts as the **brain** of an electronic project — reading inputs from sensors and buttons, then controlling outputs like LEDs, motors, and buzzers.

It is perfect for beginners because the programming language is simple and the hardware is low-cost and forgiving. With Arduino you can make things light up, move, beep, and respond to the world around them!

Input Examples	Arduino Uno Board	Output Examples
Keypad presses	ATmega328P chip	Servo motor turns
Temperature sensor	14 digital I/O pins	LED lights up
Push buttons	6 analogue input pins	Buzzer sounds
Light sensor	USB programming port	LCD displays text

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Today's Project — Keypad Password Lock

In this project you will build a **password-protected lock system** — exactly like the electronic keypads found on real safes, doors, and security gates. You will wire a keypad and a servo motor to an Arduino and program a secret password.

What you will BUILD	What you will LEARN
A working electronic lock that opens only when the correct password is entered on the keypad.	How a keypad matrix works, how to wire components, how to write password-checking code, and how to control a servo motor.

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Components Needed

Component	Description & Purpose	Qty
■ Arduino Uno	The main microcontroller board — the brain of the project. Reads keypad input and controls the servo.	1
■ 4×4 Keypad	A 16-button matrix keypad (0–9, A–D, *, #) for entering the password. Has 8 pins: 4 rows + 4 columns.	1

■ Servo Motor	Acts as the lock mechanism. Rotates to a specific angle when the correct password is entered to simulate opening a lock.	1
■ Jumper Wires	Male-to-male and male-to-female wires to connect the keypad, servo, and Arduino. Use colour-coded wires to stay organised.	~15
■ Breadboard	Solderless prototyping board. Use it to organise and share power and ground connections between components.	1
■ USB Cable	Type-B USB cable to connect Arduino to a computer for programming and to power the board during development.	1

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How the 4×4 Keypad Works

A keypad works like a grid — called a **matrix**. It has 4 rows and 4 columns, giving 16 buttons. Each button sits at the intersection of one row wire and one column wire.

When you press a button, it connects that row wire to that column wire. Arduino sends a signal through each row one at a time and listens on the columns. When it detects a connection, it knows exactly which button was pressed — this is called **matrix scanning**.

← Columns →	C1	C2	C3	C4
R1	1	2	3	A
R2	4	5	6	B
R3	7	8	9	C
R4	*	0	#	D

4×4 Keypad button layout — rows (R1–R4) × columns (C1–C4)

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Wiring the Keypad to Arduino

The 4×4 keypad has **8 pins** — 4 for rows (R1–R4) and 4 for columns (C1–C4). Connect them to Arduino digital pins as shown below. **Always power OFF the Arduino before connecting any wires!**

Keypad Pin	Function	Arduino Pin	Wire Colour (suggestion)
R1	Row 1	Digital Pin 9	■ Red
R2	Row 2	Digital Pin 8	■ Orange
R3	Row 3	Digital Pin 7	■ Yellow
R4	Row 4	Digital Pin 6	■ Green
C1	Column 1	Digital Pin 5	■ Blue
C2	Column 2	Digital Pin 4	■ Purple
C3	Column 3	Digital Pin 3	■ White
C4	Column 4	Digital Pin 2	■ Black

Servo Motor Wiring:

Servo Wire	Connects to
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■ Red (Power)	Arduino 5V pin
■ Brown/Black (GND)	Arduino GND pin
■ Orange/Yellow (Signal)	Arduino Digital Pin 10

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Coding Basics — How the Sketch Works

The Arduino sketch (program) has three main jobs: read the keypad, compare the input to the stored password, and control the servo motor accordingly.

Step 1 — Setup	Initialise the keypad pins and servo motor. Set the servo to the locked position (0°). Define the secret password.
Step 2 — Read Key	In the main loop, constantly check if a key has been pressed using <code>keypad.getKey()</code> . Append each pressed key to an input string.
Step 3 — Check Match	When enough digits are entered, compare <code>inputCode</code> to password using an if statement.
Step 4 — Unlock / Lock	If password matches → <code>servo.write(90)</code> to unlock. If wrong → <code>servo.write(0)</code> and clear the input.
Step 5 — Reset	After a timeout or reset key (#), clear the input string and return the servo to the locked position.

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Sample Arduino Code Walkthrough

Key Code Lines Explained

<code>#include #include</code>	Include the Keypad and Servo libraries — pre-written code that makes the hardware easy to use.
<code>char password[] = "1234";</code>	Store the secret password as a character array. Change "1234" to your own password!
<code>char key = keypad.getKey();</code>	Read which button is currently being pressed. Returns 0 if no key is pressed.
<code>inputCode += key;</code>	Append the pressed character to the input string, building up the entered password.
<code>if (inputCode == password)</code>	Compare the entered string to the stored password. If they match, unlock!
<code>servo.write(90);</code>	Rotate the servo to 90 degrees — this opens the lock mechanism.
<code>servo.write(0);</code>	Rotate the servo back to 0 degrees — this closes/locks the mechanism.
<code>inputCode = "";</code>	Clear the input string so the user can try again.

Complete Sketch Outline

```
#include

#include

const byte ROWS = 4, COLS = 4;
char keys[ROWS][COLS] = {'1','2','3','A'},
{'4','5','6','B'},
{'7','8','9','C'},
{'*','0','#','D'};

byte rowPins[ROWS] = {9,8,7,6};
byte colPins[COLS] = {5,4,3,2};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

Servo servo;

String password = "1234";
String inputCode = "";

void setup() {
  servo.attach(10);
  servo.write(0); // start locked
}

void loop() {
  char key = keypad.getKey();
  if (key) {
    inputCode += key;
    if (inputCode.length() == password.length()) {
      if (inputCode == password) {
        servo.write(90); // UNLOCK
        delay(3000);
        servo.write(0); // lock again
      }
      inputCode = ""; // reset input
    }
  }
}
```

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Testing Your Project

Follow these steps carefully to test your password lock for the first time:

1	Upload the sketch to Arduino via the Arduino IDE. Check for any error messages.
2	Connect the Arduino to power (USB cable).
3	Verify the servo is in the locked position (0°) at startup.
4	Press the correct password on the keypad (e.g., 1-2-3-4).
5	Watch the servo rotate to 90° — the lock opens!
6	Wait 3 seconds — the servo returns to 0° (locks again).
7	Try entering a WRONG password. The servo should NOT move.
8	Try changing the password in the code and re-upload to test.

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Troubleshooting Common Problems

■■ Keypad not responding	Double-check that pin numbers in your code match the physical wiring. Ensure the Keypad library is installed in Arduino IDE.
■■ Servo does not move	Test the servo alone with a simple sweep sketch first. Check that the servo signal wire is on pin 10 and power/GND are correct.
■ Correct password not unlocking	Check that inputCode and password are the same length and type. Print both to Serial Monitor to compare: <code>Serial.println(inputCode);</code>
■ Wrong passwords unlocking	Ensure the <code>==</code> comparison is used (not <code>=</code>). Check there are no extra spaces or newline characters in the password string.
■ Code upload fails	Disconnect the keypad TX/RX pins (0, 1) before uploading — they conflict with USB serial communication.
■ Arduino not powering on	Try a different USB cable or port. Check the Arduino power LED (green) — if off, the board may need a driver reinstall.

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Safety Tips

■ Correct Voltage Only	Only power Arduino with USB (5V) or recommended batteries. Never use a higher voltage — it will permanently damage the board.
■ Keep Liquids Away	Water and electronics do not mix! Keep your workspace dry and away from drinks.
■ Handle with Care	Pick up components gently. Avoid bending pins or dropping the Arduino. Static electricity can damage chips.
■ Power Off Before Wiring	Always unplug the Arduino before connecting or changing any wires. Never wire components while powered on.
■■■ Ask for Help	If something looks wrong or you are unsure, ask your teacher before powering on. It is always safe to check first!

11 Real-World Applications

Keypad password locks are used all around the world. Your project today is the same basic technology behind these real systems:

Application	How it connects to your project
■ Office Door Locks	Offices, schools, and government buildings use electronic keypads to restrict access. Only authorised people with the code can enter.
■ Home Safes	Home and bank safes use keypad combinations to protect valuables. The same servo+keypad logic you built controls real locking bolts.
■ Car Alarms & Gates	Driveways and secure car parks use keypad entry systems to allow authorised vehicles and people through gates.
■ ATM Machines	ATM PIN pads use matrix keypad technology to securely read your 4-digit PIN — just like your project reads the password.
■ Phone Lock Screens	Your smartphone PIN lock works on the same principle: compare entered digits to stored password, grant access if correct.
■ Industrial Control	Factories use keypad-controlled panels to authorise the operation of dangerous machinery or restricted areas.

12 Summary — What You Learned Today

■ Arduino Basics	Arduino is a programmable microcontroller that reads inputs and controls outputs.
■■ Keypad Matrix	A 4×4 keypad uses 8 pins and matrix scanning to detect 16 different button presses.
■ Wiring	Keypad R1–R4 → Pins 9–6; C1–C4 → Pins 5–2; Servo → Pin 10 (signal), 5V, GND.
■ Code Logic	Setup servo and keypad → read key presses → build input string → compare to password → control servo.
■ Password Checking	if (inputCode == password) unlocks; servo.write(90) opens, servo.write(0) closes.
■■ Safety	Use correct voltage, power off before wiring, handle gently, and keep water away.
■ Real World	This same technology is in office keypads, bank safes, ATMs, and car gate systems.

13 Quiz Time! ■

1	What does Arduino do in this project? A) It stores files B) It reads keypad input and controls the servo C) It connects to the internet
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How many pins does the 4×4 keypad have, and what are they for?

A) 4 pins — one per row B) 16 pins — one per button C) 8 pins — 4 rows + 4 columns

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What happens if you enter the WRONG password?

A) The servo rotates to 90° B) The input clears and you try again C) The Arduino resets

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Which Arduino function reads a pressed key?

A) keypad.press() B) keypad.getKey() C) Serial.read()

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What angle does servo.write(90) rotate the servo to, and what does it do?

A) 0° — locks B) 45° — halfway C) 90° — unlocks

*Answers: 1→B, 2→C, 3→B, 4→B, 5→C***■ Great Work — Keep Building & Experimenting!**